

Harbor North 138 kV Substation and Feeder Extension: Design Proposal

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Design proposal for a new Harbor North 138 kV substation and feeder extension to serve industrial and port load growth in the north part of Harbor District. Prepared by Distribution Planning for the capital committee, September 3, 2026.

1. Proposal

Build a new 138 kV substation with two 80 MVA transformers and four 34.5 kV feeders. Estimated cost is \$85 million, with \$6.5 million budgeted in 2026 for design and long-lead equipment. Planned in-service date: December 15, 2028. Because the project exceeds \$50 million it needs board approval.

2. Need

Harbor is the port and industrial corridor and accounted for 48.2% of the 25.1 TWh delivered in 2025, with 128,640 of the 684,930 delivery points at the end of August 2026.

At August 31, 2026 there were 11 large-load requests of 20 MW or more totaling 1,905 MW, including one withdrawn 120 MW request; the 10 active requests total 1,785 MW. Six of the requests, 1,160 MW, are in Harbor, and one 85 MW port-electrification request already has an executed agreement.

3. Large-load requests

The table totals requested megawatts and security posted for all 11 requests, including the withdrawn request. In June 2026 the PUCT approved ERCOT's batch study process for large-load interconnection requests, which will set the order in which the largest requests are studied.

Measure	Value
Measure	requested_mw
Unit	MW
Total	1905.0
Mean	173.1818
Records	11
Period	2026-08-31 to 2026-08-31
Measure	security_posted_usd
Unit	USD
Total	17650000.0

Mean	1604545.4545
Records	11
Period	2026-08-31 to 2026-08-31

Large-load interconnection requests of 20 MW or more that were open or withdrawn at August 31, 2026, with requested MW, security posted, load type, requested energization month and study stage.

4. Planning criteria

No overload with one transformer or feeder out of service (N-1), and substation loading below 80% of nameplate at forecast peak.

5. Alternatives

Expanding two existing Harbor substations would cost less but would not provide enough capacity for requests beyond 2028. Serving the largest requests directly from the transmission system remains an option for individual customers but does not address distribution-level growth.

6. Open issue

The load forecast uses requested megawatts rather than probability-weighted load. Distribution Planning has flagged this for the capital committee: if only the requests at facilities study or later proceed, the transformer size and the in-service date should be reviewed.

7. Request

Approve continued design and the long-lead equipment commitment within the \$6.5 million 2026 budget, with a full funding request to the board after the load forecast is re-run.